BIRZEIT UNIVERSITY

Faculty of Engineering and Technology
Electrical and Computer Engineering Department
ENCS4130 – Computer Network Laboratory

Experiment No. 10

Internet Protocol Version 6 (IPv6) Configuration

■ Objectives

(a) Learn how to configure IPv6 addressing.

(b) Understand static IPv6 routing.

(c) Understand dynamic IPv6 routing using RIPng.

■ Requirements

(a) Cisco Packet Tracer software.

(b) Five routers.

(c) Three switches.

(d) Four PCs, two servers (Web and DNS), and one printer.

■ Pre-Lab Task

Students are required to build the network topology shown in Figure 2, including all notes, following the guidelines provided in Section 10.2.1. In addition, they must complete all IPv4 configurations (IP addressing and static routing) as described in Section 10.2.2.

10.1 Introduction

You have likely heard that the world is running out of IPv4 addresses. According to the Number Resource Organization (NRO), as of February 3, 2011, the global pool of unallocated IPv4 addresses was officially exhausted. IPv6 was introduced to overcome this limitation. IPv6 addresses are 128 bits long, providing approximately 3.4×10^{38} unique addresses.

Unlike IPv4, IPv6 addresses are written in *hexadecimal* and separated by *colons*. Each group consists of *four hexadecimal digits*, and leading zeros within a group may be omitted. For example:

AA76:0000:0000:0000:0012:A322:FE33:2267

can be shortened to:

AA76:0:0:0:12:A322:FE33:2267

Additionally, *one* consecutive sequence of zero groups can be replaced by a double colon (**::**) for further simplification:

AA76::12:A322:FE33:2267

Important:

A double colon (**::**) may appear *only once* in an IPv6 address. For example:
AA76:0000:0000:0012:A322:0000:0000:2267
can be written as either:
AA76::12:A322:0:0:2267
or:
AA76:0:0:12:A322::2267
but **not**:
AA76::12:A322::2267
Using more than one double colon in a single IPv6 address is invalid.

10.1.1 IPv6 Address Types

IPv6 supports several types of addresses, each serving a specific purpose in communication.

- **Unicast:** A unicast address *identifies one single interface.* Packets sent to a unicast address are *delivered to exactly one destination.*
  - **Global Unicast Addresses:** These are globally routable addresses, similar in function to public IPv4 addresses. They fall within the prefix **2000::/3**.
  - **Link-Local Addresses:** These are used for communication on the same local link only and are not routable beyond that link. All IPv6-enabled interfaces automatically generate a link-local address. They fall within **FE80::/10**. Link-local addresses are essential for functions such as Neighbor Discovery and Stateless Address Autoconfiguration (SLAAC).
- **Multicast:** A multicast address *identifies a group of interfaces*, typically on different devices. Packets sent to a multicast address are *delivered to all members of*

the group. In IPv6, multicast *replaces broadcast*, which does not exist in IPv6.

- **Anycast:** An anycast address is *assigned to multiple interfaces*, usually on different devices. Packets sent to an anycast address are *delivered to the nearest interface* (based on routing metrics). Anycast addresses are usually taken from the unicast address space, not from a special reserved block.
- **Reserved / Special Addresses:** IPv6 defines several special-purpose addresses:
  - **::1 (Loopback Address):** Used by a host to send packets to itself.
  - **:: (Unspecified Address):** Indicates the absence of an address (e.g., before a device obtains an IPv6 address). It cannot be assigned to any interface.
  - **IPv4-mapped IPv6 Addresses:** These addresses embed an IPv4 address within an IPv6 address (e.g., **::FFFF:192.0.2.1**). They allow IPv6-only applications to communicate with IPv4 systems by representing IPv4 destinations in IPv6 format.

10.1.2 Tunneling

During the transition from IPv4 to IPv6, a major challenge is that much of the current network infrastructure still operates using IPv4. Since a full migration to IPv6 cannot happen instantly, mechanisms are required to enable IPv6 networks to communicate across IPv4-only segments. One of the most widely used transition techniques is **tunneling**.

Tunneling works by *encapsulating an IPv6 packet inside an IPv4 packet*. The added IPv4 header allows the packet to travel across IPv4-only parts of the network, while the original IPv6 packet remains unchanged. Once the encapsulated packet reaches the other end of the tunnel, the IPv4 header is removed (**decapsulation**), and the IPv6 packet is forwarded normally. This effectively creates a *virtual point-to-point IPv6 link* between two IPv6 routers, even if the physical path supports only IPv4, as illustrated in Figure 1.

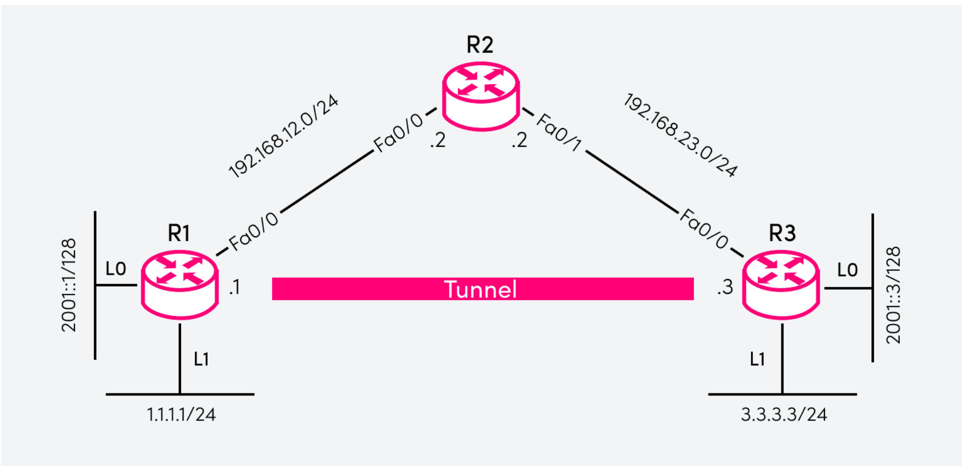


Figure 1: Encapsulation of IPv6 packets inside IPv4 headers in a 6in4 tunnel.

One well-known technique for configuring a tunnel is the **Manual IPv6-over-IPv4 Tunnel (6in4)**. In a manually configured 6in4 tunnel, each router is explicitly configured with a tunnel interface.

- The **tunnel source** is the IPv4 address of the local router.
- The **tunnel destination** is the IPv4 address of the remote router.
- Each tunnel interface is assigned an IPv6 address, usually taken from the **same IPv6 subnet**, allowing the routers to appear directly connected at Layer 3.

This setup allows IPv6 routing protocols—such as *RIPng*, *OSPFv3*, or *static IPv6 routes*—to operate across the tunnel, even though the underlying infrastructure is IPv4-only.

Tunneling is widely used as a transition mechanism because it provides IPv6 connectivity without requiring immediate IPv6 support across the entire network. However, it is considered a *temporary solution*. The long-term objective is the full *native IPv6 deployment*, which eliminates the need for tunneling and enables true end-to-end IPv6 communication.

10.1.3 Configuring Cisco Routers with IPv6

A) Configuring a Router Interface

To assign an IPv6 address to an interface:

```
Router(config)# interface <type><slot>/<port>
Router(config-if)# ipv6 address <ipv6-prefix>/<prefix-length>
```

Important:

An interface can have **multiple** IPv6 addresses. It may also be configured with both IPv4 and IPv6 simultaneously (dual stack).

B) Enabling IPv6 Routing

IPv6 routing is *disabled by default* on Cisco routers. To enable it:

```
Router(config)# ipv6 unicast-routing
```

C) Static IPv6 Routing

To configure a static IPv6 route:

```
Router(config)# ipv6 route <ipv6-prefix>/<prefix-length>
                <next-hop-ipv6-address>
```

- `ipv6 route`: Command to create a static IPv6 route.
- `<ipv6-prefix>/<prefix-length>`: The destination network and its prefix length.
- `<next-hop-ipv6-address>`: The IPv6 next-hop router on a directly connected link.

This instructs the router to forward packets matching `<ipv6-prefix>/<prefix-length>` to the specified next-hop IPv6 address.

D) Dynamic IPv6 Routing: RIPng

Routing Information Protocol next generation (RIPng) is the IPv6 version of RIPv2. Key characteristics include:

- Distance-vector routing protocol
- Maximum hop count: 15
- Uses *link-local addresses* as next-hop addresses
- Enabled *per interface*, not with a global **network** statement

To enable RIPng:

```
Router(config)# interface <type><slot>/<port>
Router(config-if)# ipv6 rip <rip-id> enable
```

The `<rip-id>` identifies the RIPng routing process (can be numeric or textual).

E) Configuring IPv6 Tunneling

Tunneling enables IPv6 packets to travel across IPv4-only infrastructure by encapsulating IPv6 packets inside IPv4 headers. To configure a manual IPv6-over-IPv4 tunnel (**6in4**) on Cisco routers, use the following commands:

```
Router(config)# interface tunnel <tunnel-number>
Router(config-if)# ipv6 address <ipv6-prefix>/<prefix-length>
Router(config-if)# tunnel source <ipv4-source-interface-or-
                   address>
Router(config-if)# tunnel destination <ipv4-destination-
                   address>
Router(config-if)# tunnel mode ipv6ip
```

- `interface tunnel`: Creates a virtual tunnel interface.
- `ipv6 address`: Assigns an IPv6 address to the tunnel.
- `tunnel source`: Defines the local IPv4 endpoint (interface or IP address).
- `tunnel destination`: Defines the remote IPv4 endpoint of the tunnel.
- `tunnel mode ipv6ip`: Specifies an IPv6-over-IPv4 encapsulation tunnel.

The IPv6 packet is encapsulated at the source router, transported across IPv4-only links, and decapsulated at the remote tunnel endpoint.

> **Important:**
>
> Both ends of the tunnel must be configured consistently. IPv6 routing protocols (such as RIPng or OSPFv3) can run over the tunnel to exchange routes.

F) Configuring DHCPv6

To configure a DHCPv6 pool that provides IPv6 configuration parameters (such as DNS server and network prefix) to clients:

```
Router(config)# ipv6 dhcp pool <pool-name>
Router(config-dhcp)#address prefix <ipv6-prefix>/<prefix-len>
Router(config-dhcp)# dns-server <dns-ipv6-address>
```

Next, bind the pool to the interface and enable IPv6 router advertisements with the appropriate flags:

```
Router(config)# interface <type><slot>/<port>
Router(config-if)# ipv6 nd other-config-flag
Router(config-if)# ipv6 dhcp server <pool-name>
```

- `ipv6 nd other-config-flag`: Sets the"Other Configuration" (O) flag in router advertisements, indicating that clients should use DHCPv6 for additional configuration (such as DNS).

H) DNS "AAAA" Records

An `AAAA` record maps a hostname to an IPv6 address. It provides IPv6-enabled name resolution and is the IPv6 equivalent of an `A` record for IPv4.

10.1.4 Cisco Discovery Protocol (CDP)

CDP is a proprietary *Layer 2 protocol* developed by Cisco Systems. It allows Cisco devices to automatically discover and exchange information with directly connected Cisco devices. Since CDP operates at Layer 2, it functions independently of Layer 3 protocols, including IPv4 and IPv6. CDP is commonly used for *network management*, *troubleshooting*, and *device inventory*.

CDP-enabled devices *periodically send advertisement messages* to a *multicast MAC address* (`01:00:0C:CC:CC:CC`). These messages contain useful information such as the device ID (hostname), IP address, operating system version, hardware platform (e.g., router or switch model), active interfaces, and device capabilities (e.g., router, switch, IP phone). Neighboring CDP-enabled Cisco devices receive these messages, extract the information, and store it in a local CDP cache.

A) Monitoring CDP

Network administrators can view CDP information using the following commands:

- Display a summary of directly connected Cisco devices:

  ```
  Router# show cdp neighbors
  ```

- Display detailed information about each neighboring device:

  ```
  Router# show cdp neighbors detail
  ```

B) Disabling CDP

CDP is enabled by default on most Cisco interfaces, but it can be disabled when not required—either per interface or globally.

- Disable CDP on a specific interface:

```
Router(config)# interface <type><slot>/<port>
Router(config-if)# no cdp enable
```

- Disable CDP globally:

```
Router(config)# no cdp run
```

Disabling CDP on interfaces connected to untrusted networks is recommended, as it prevents the disclosure of device information that could potentially be exploited by attackers.

10.2 Procedure

In this lab, we will connect five routers, three switches, and several end devices across multiple networks using both IPv4 and IPv6 addressing. This requires configuring routing protocols between the routers. For IPv6, we will use both static routing and dynamic routing (RIPng), while for IPv4 we will configure static routing.

The topology includes a DNS server and a Web server. The DNS server will be configured with both an `A` record (IPv4) and an `AAAA` record (IPv6) for the Web server, enabling name resolution over both protocols. The end devices in the network `2001:11AA::/64` (PC0 and PC1) will receive their IPv6 configuration dynamically via DHCPv6, provided by Router0.

Because the links Router1–Router2 and Router2–Router3 support only IPv4, native IPv6 traffic cannot traverse this part of the network. To maintain end-to-end IPv6 connectivity, we will implement IPv6 tunneling over IPv4. Specifically, an IPv6 tunnel will be configured between Router1 and Router3, enabling IPv6 packets to pass through the IPv4-only segment by encapsulating them inside IPv4 headers. This ensures full IPv6 connectivity across the entire topology despite the intermediate IPv4-only links.

10.2.1 Building the Topology

Construct the network topology shown in Figure 2, which consists of the following devices:

- Routers: Use **2811 IOS 15**.
- Switches: Use **Switch-PT**.
- PCs, Servers, and Printer: From "*End Devices*", use **PC-PT**, **Server-PT**, and **Printer-PT**.
- Connections: Use the "**Automatically Choose Connection Type**" option to connect devices.

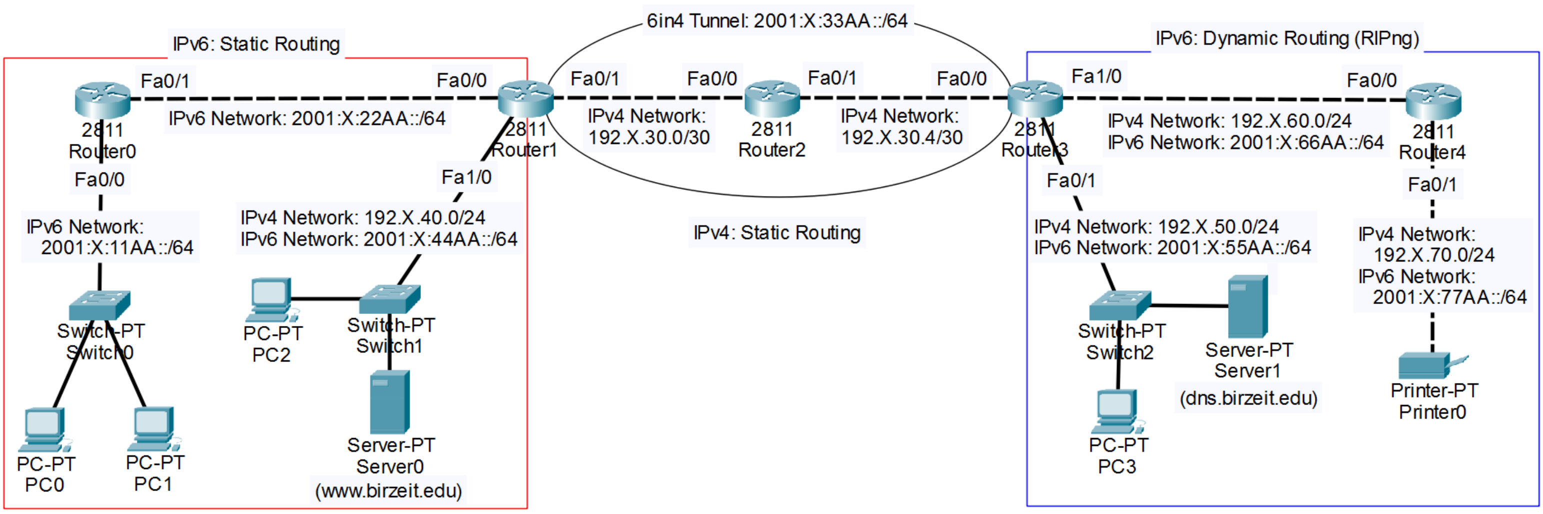


Figure 2: IPv6 topology.

For **Router1** and **Router3**, an additional interface is required. To install it:

1. Click the router and open the ***Physical*** tab.

2. Turn off the router using the power switch.
3. Insert the ***NM-1FE2W*** module into any empty slot.
4. Turn the router back on.

10.2.2 IPv4 Configuration

This section covers the IPv4 configurations, including addressing and static routing.

A) Router Interface Configuration

To configure the IPv4 address on interface `FastEthernet1/0` of **Router1**, use the commands below:

```
Router(config)# interface Fa1/0
Router(config-if)# ip address 192.X.40.1 255.255.255.0
Router(config-if)# no shutdown
```

Use the same procedure to configure the remaining interfaces: **Router1** (interface `Fa0/1`), **Router2** (interfaces `Fa0/0` and `Fa0/1`), **Router3** (interfaces `Fa0/0`, `Fa0/1`, and `Fa1/0`), and **Router4** (interfaces `Fa0/0` and `Fa0/1`). Assign each interface the correct IPv4 address and subnet mask as shown in the topology diagram.

B) Routing Configuration

All IPv4 networks use ***static routing*** to reach remote networks. To configure static routes on **Router1**, use:

```
Router(config)# ip route 192.X.30.4 255.255.255.252
                192.X.30.2
Router(config)# ip route 192.X.50.0 255.255.255.0 192.X.30.2
Router(config)# ip route 192.X.60.0 255.255.255.0 192.X.30.2
Router(config)# ip route 192.X.70.0 255.255.255.0 192.X.30.2
```

Repeat the process to configure static routes on **Router2**, **Router3**, and **Router4**, using the appropriate network addresses and next-hop IPs.

C) Server Configuration

Configure both the Web Server and DNS Server as described in **Experiment No. 6**.

- Web Server (`www.birzeit.edu`)
  - Assign a static IPv4 configuration, as shown in Figure 3.
  - Enable HTTP and HTTPS (Secure HTTP) protocols.
  - Customize the `index.html` page to display the title: "**Birzeit University**".
- DNS Server (`dns.birzeit.edu`)
  - Assign a static IPv4 configuration.
  - Enable only the DNS service.

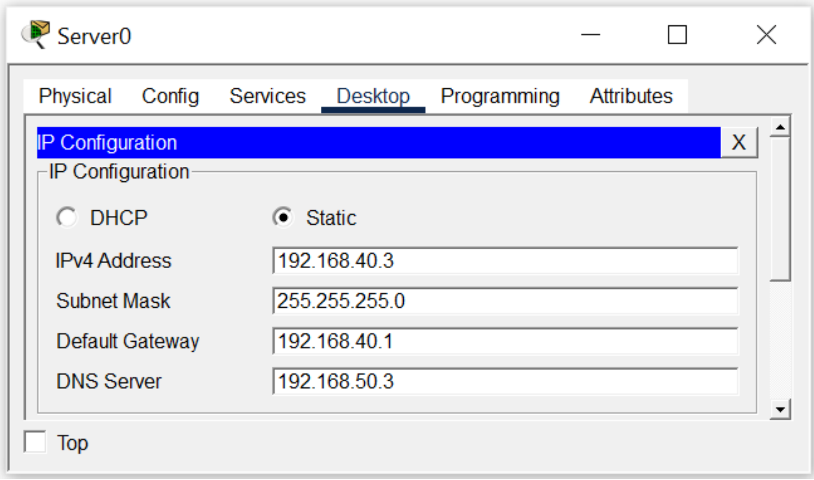


Figure 3: IPv4 configuration of Birzeit Web server.

– Add the resource record listed in Table 1, as illustrated in Figure 4.

Table 1: Resource record (IPv4) in Birzeit DNS server.

Type	**Name**	**Value**
A	`www.birzeit.edu`	The IP address of the `www.birzeit.edu` server

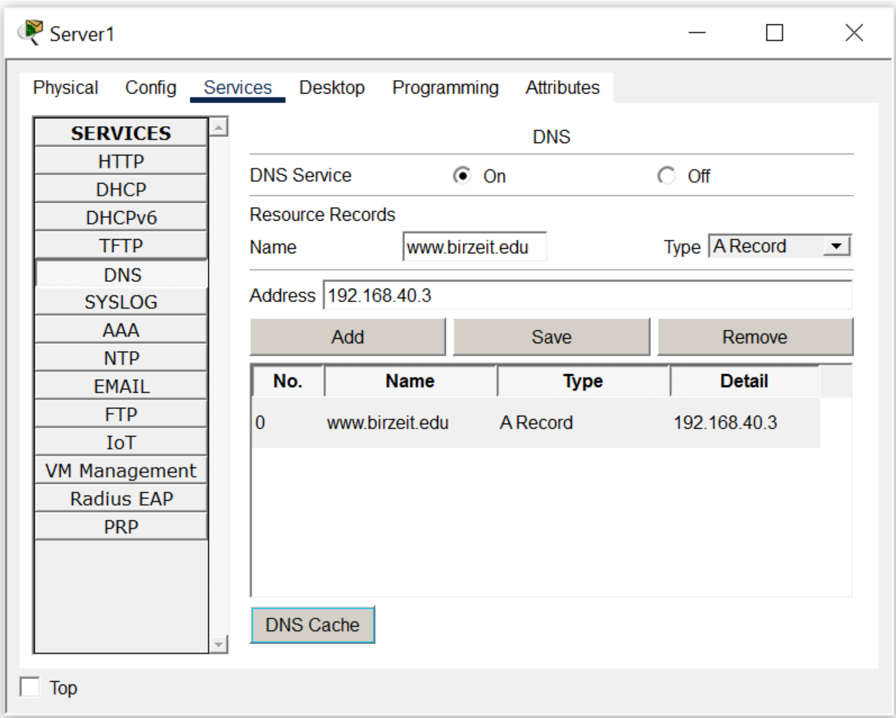


Figure 4: DNS service configuration (IPv4 Resource record) of Birzeit DNS server.

D) End Device Configuration

Configure IPv4 settings for **PC2**, **PC3**, and **Printer0**, including IPv4 address, subnet mask, default gateway, and IPv4 address of the DNS server. Follow the same procedure used in previous experiments.

10.2.3 IPv6 Configuration

This section covers the IPv6 configuration tasks, including addressing, static routing, dynamic routing (RIPng), and DHCPv6.

A) Router Interface Configuration

To configure the IPv6 address on interface `FastEthernet1/0` of **Router1**, use:

```
Router(config)# interface Fa1/0
Router(config-if)# ipv6 address 2001:X:44AA::1/64
Router(config-if)# no shutdown
```

> Note:
>
> Interface `FastEthernet1/0` on **Router1** is now configured with both IPv4 and IPv6 addresses (dual-stack operation).

Use the same procedure to configure the remaining IPv6-enabled interfaces: **Router0** (interfaces `Fa0/0` and `Fa0/1`), **Router1** (interface `Fa0/0`), **Router3** (interfaces `Fa0/1` and `Fa1/0`), and **Router4** (interfaces `Fa0/0` and `Fa0/1`). Assign each interface the appropriate IPv6 prefix and prefix length as shown in the topology diagram.

B) Routing Configuration

IPv6 routing requires two mechanisms across the topology:

- Static routing on **Router0** and **Router1**.
- RIPng dynamic routing on **Router3** and **Router4**.
- Additional static routes on **Router3** and **Router4** for networks not advertised by RIPng.

Accordingly, **Router0** and **Router1** should use *static* IPv6 routes to reach remote networks. To enable IPv6 routing and configure static routes on **Router1**:

```
Router(config)# ipv6 unicast-routing
Router(config)# ipv6 route 2001:X:11AA::/64 2001:X:22AA::1
Router(config)# ipv6 route 2001:X:55AA::/64 2001:X:33AA::2
Router(config)# ipv6 route 2001:X:66AA::/64 2001:X:33AA::2
Router(config)# ipv6 route 2001:X:77AA::/64 2001:X:33AA::2
```

Similarly, configure the appropriate static routes on **Router0** according to the topology.

On the other hand, **Router3** and **Router4** exchange IPv6 routing information dynamically using *RIPng* for the following networks: `2001:X:55AA::/64`, `2001:X:66AA::/64` and `2001:X:77AA::/64`. Since **Router3** and **Router4** do not learn the remaining IPv6 networks (i.e., `2001:X:11AA::/64`, `2001:X:22AA::/64`, and `2001:X:44AA::/64`) through RIPng, these networks must be added *statically* to ensure full IPv6 connectivity.

To enable IPv6 routing on **Router3** and apply the required static and RIPng configurations, use:

```
Router(config)# ipv6 unicast-routing
Router(config)# interface Fa0/1
Router(config-if)# ipv6 rip 1 enable
Router(config-if)# exit
Router(config)# interface Fa1/0
Router(config-if)# ipv6 rip 1 enable
Router(config-if)# exit
Router(config)# ipv6 route 2001:X:22AA::/64 2001:X:33AA::1
Router(config)# ipv6 route 2001:X:11AA::/64 2001:X:33AA::1
Router(config)# ipv6 route 2001:X:44AA::/64 2001:X:33AA::1
```

Configure **Router4** similarly, enabling RIPng on its participating interfaces and adding static routes for IPv6 networks outside the scope of RIPng.

C) DHCPv6 Pool Configuration

This configuration defines a DHCPv6 pool on **Router0** that provides IPv6 clients in the `2001:X:11AA::/64` network with configuration parameters such as the IPv6 address, prefix length, default gateway, and DNS server address.

```
Router(config)# ipv6 dhcp pool POOL0
Router(config-dhcp)# address prefix 2001:X:11AA::/64
Router(config-dhcp)# dns-server 2001:X:55AA::3
```

Then, the pool must be bound to interface `FastEthernet0 /0`. The "`other-config-flag`" instructs hosts to use DHCPv6 for obtaining extra configuration options (such as DNS) while still using SLAAC for address autoconfiguration.

```
Router(config)# interface Fa0/0
Router(config-if)# ipv6 nd other-config-flag
Router(config-if)# ipv6 dhcp server POOL0
```

D) Server Configuration

Configure the IPv6 addresses for the Web Server and DNS Server as described below.

- Web Server (`www.birzeit.edu`)
  - Assign a static IPv6 configuration, as shown in Figure 5.
- DNS Server (`dns.birzeit.edu`)
  - Assign a static IPv6 configuration.
  - Add the resource record listed in Table 2, as illustrated in Figure 6.

Table 2: Resource record (IPv6) in Birzeit DNS server.

Type	Name	Value
AAAA	`www.birzeit.edu`	The IPv6 address of the `www.birzeit.edu` server

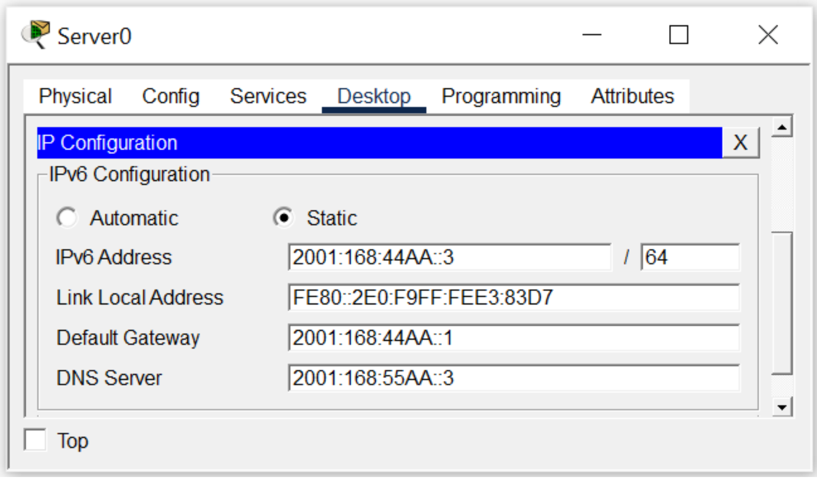


Figure 5: IPv6 configuration of Birzeit Web server.

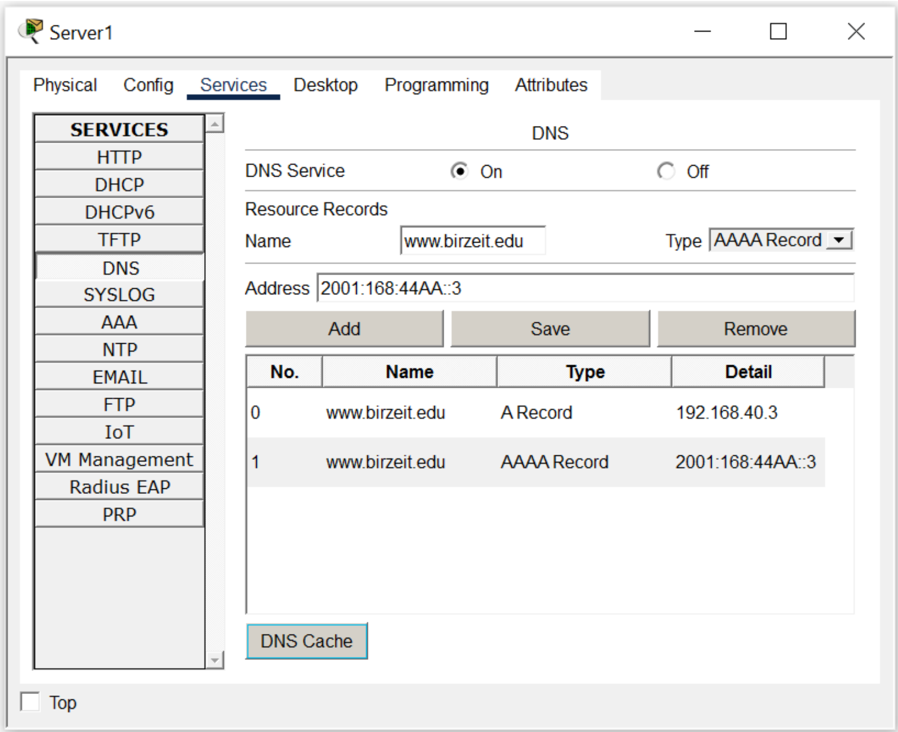


Figure 6: DNS service configuration (IPv6 Resource record) of Birzeit DNS server.

E) End Device Configuration

Configure IPv6 settings ***statically*** for **PC2**, **PC3**, and **Printer0**, including IPv6 address, prefix length, default gateway, and IPv6 address of the DNS server. Follow the same configuration steps used earlier for the servers.

In contrast, **PC0** and **PC1** should receive their IPv6 configuration ***dynamically*** using DHCPv6. To configure these devices:

- Click on the PC to open its configuration window.
- Navigate to the ***Desktop*** tab and select ***IPv6 Configuration***.

- Set the IPv6 address configuration method to ***Automatic***, as shown in Figure 7.

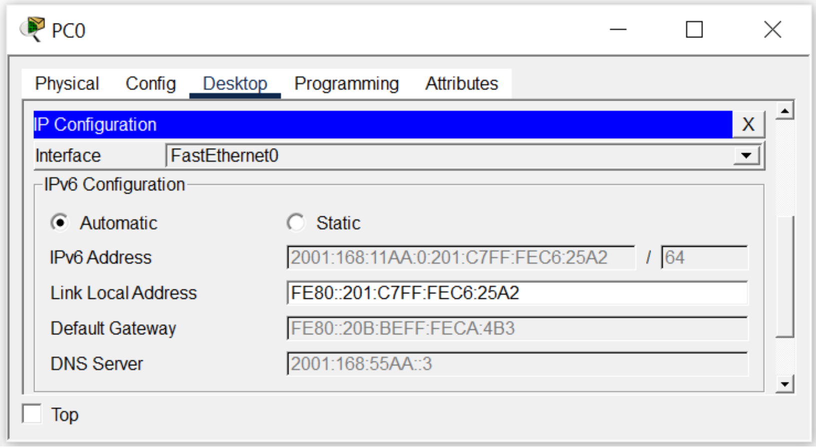


Figure 7: IPv6 configuration of PC0.

10.2.4 Configuring IPv6 Tunneling Between Router1 and Router3

Because the physical path between **Router1** and **Router3** operates using IPv4 only (through **Router2**), an IPv6-over-IPv4 tunnel is required. This tunnel creates a virtual point-to-point IPv6 link directly between **Router1** and **Router3**. IPv6 packets are encapsulated inside IPv4 headers so they can traverse the IPv4-only segment.

Use the following commands to configure the tunnel interface on **Router1** with the IPv6 tunnel network `2001:X:33AA::/64`:

```
Router(config)# interface tunnel 1
Router(config-if)# ipv6 address 2001:X:33AA::1/64
Router(config-if)# tunnel source FastEthernet0/1
Router(config-if)# tunnel destination 192.X.30.6
Router(config-if)# tunnel mode ipv6ip
```

In this configuration:

- `Tunnel1` is created on **Router1** and assigned an IPv6 address that represents one endpoint of the virtual tunnel.
- The tunnel source is the IPv4 address assigned to **Router1**'s `FastEthernet0/1` interface.
- The tunnel destination is the IPv4 address of **Router3**'s `FastEthernet0/0` interface.
- The `ipv6ip` mode specifies that IPv6 packets will be encapsulated inside IPv4 packets.

Similarly, configure the tunnel interface on **Router3**, assigning it the corresponding tunnel IPv6 address and setting the correct tunnel source and destination IPv4 addresses.

10.2.5 Displaying CDP Neighbor Information

To view information about directly connected Cisco devices, use:

```
Router# show cdp neighbors
```

This command displays a summary of neighboring devices, including device IDs, local interfaces, capabilities, platforms, and connected ports as shown in Figure 8.

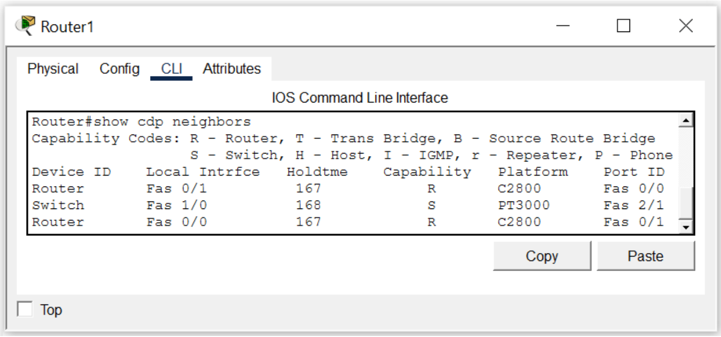


Figure 8: Output of the `show cdp neighbors` command.

For more detailed information—such as IP addresses, software versions, and interface settings—use:

```
Router# show cdp neighbors detail
```

The detailed output (Figure 9) is useful for troubleshooting and device documentation.

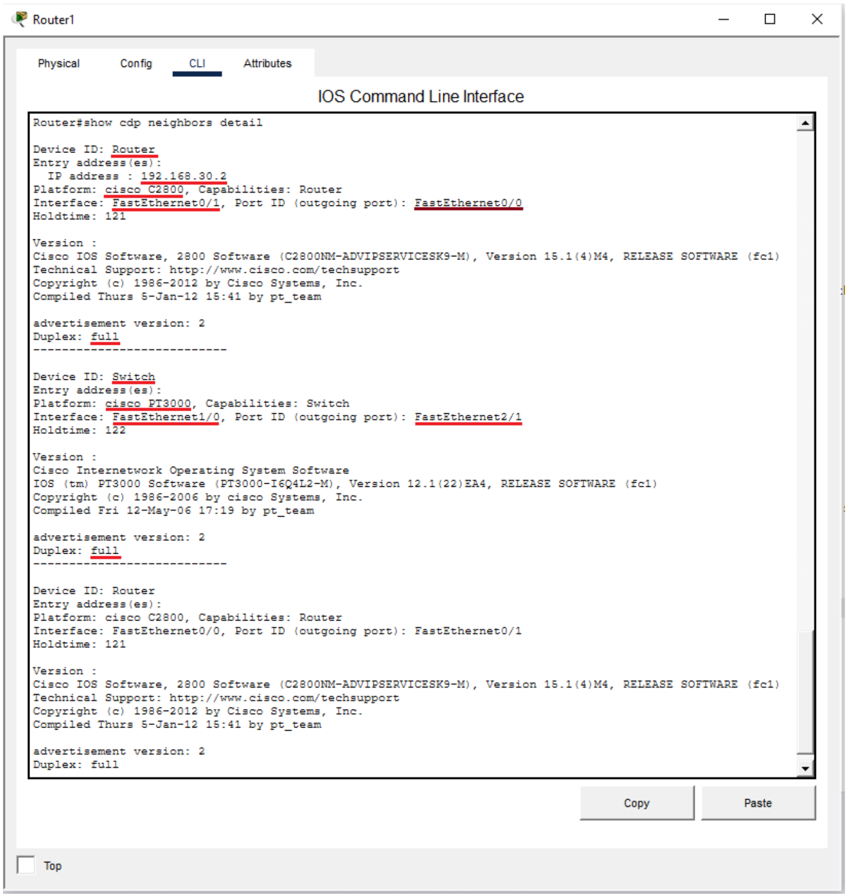


Figure 9: Output of the `show cdp neighbors detail` command.

10.3 Verifying Your Configuration

A) Ping and Traceroute from PC0 to Remote IPv6 Networks

To verify end-to-end IPv6 connectivity, **PC0** generates *ICMPv6 Echo Request* messages and sends them to devices located in remote IPv6 networks across the topology.

Figure 10 demonstrates a successful IPv6 ping from **PC0** to multiple remote IPv6 destinations. The receipt of *Echo Reply* messages for all requests confirms that the IPv6-over-IPv4 tunnel established between **Router1** and **Router3** is functioning correctly, and that IPv6 routing is properly configured across all network segments.

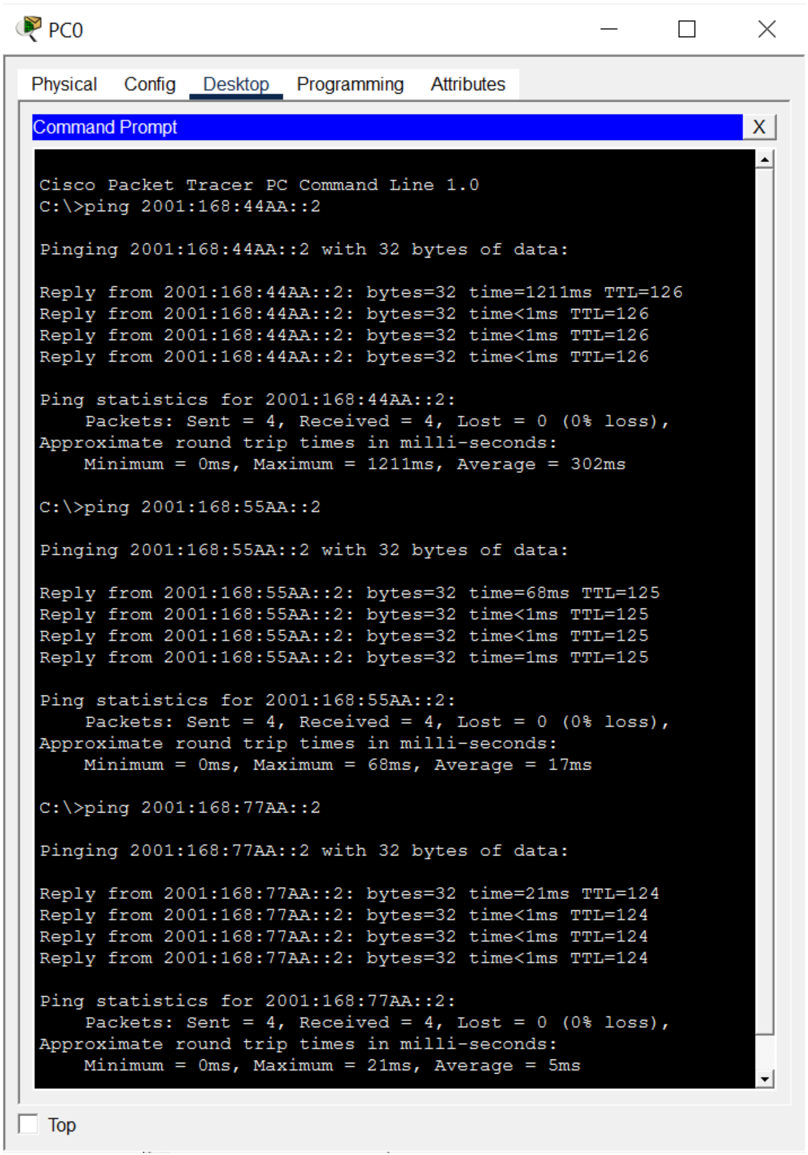


Figure 10: Successful IPv6 ping from PC0 to remote IPv6 networks.

Figure 11 illustrates the packet encapsulation process at **Router1**. To traverse the IPv4-only segment of the network, **Router1** encapsulates the original IPv6 packet (from **PC0** to **Printer0**) within an IPv4 packet whose destination address is **Router3**'s IPv4 tunnel endpoint. The encapsulated packet is forwarded toward **Router3** via **Router2**, enabling IPv6 traffic to be transported transparently across the intermediate IPv4 infrastructure.

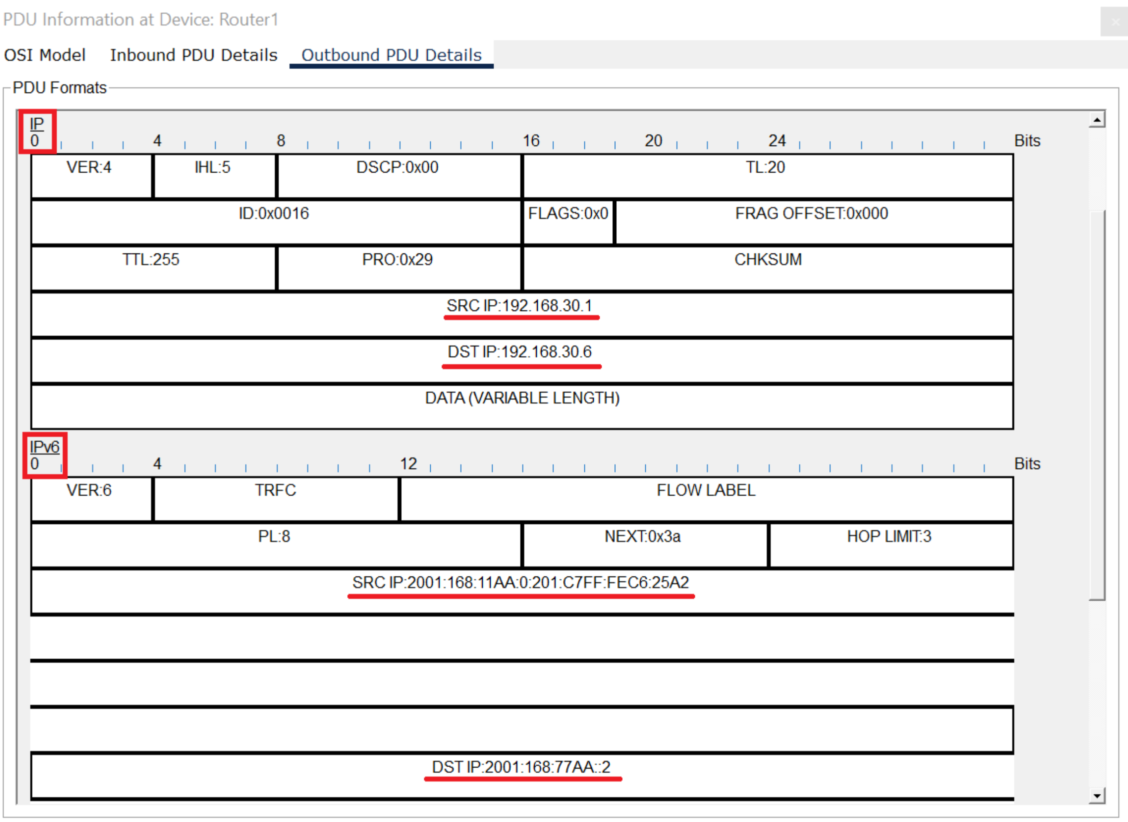


Figure 11: IPv6 packet encapsulated within an IPv4 packet at Router1.

Figure 12 shows the output of a traceroute from **PC0** to **Printer0** (`2001:X:77AA::2`). The traceroute reveals that IPv6 packets first reach **Router0** (`2001:X:11AA::1`), followed by **Router1** (`2001:X:22AA::2`). The next hop is **Router3**'s IPv6 tunnel endpoint (`2001:X:33AA::2`), indicating successful traversal of the IPv6-over-IPv4 tunnel across the IPv4-only segment. After decapsulation at **Router3**, the packets are forwarded to **Router4** (`2001:X:66AA::2`) and finally delivered to the destination, **Printer0** (`2001:X:77AA::2`). This hop-by-hop path confirms the correct operation of the IPv6-over-IPv4 tunneling mechanism and proper IPv6 routing beyond the tunnel.

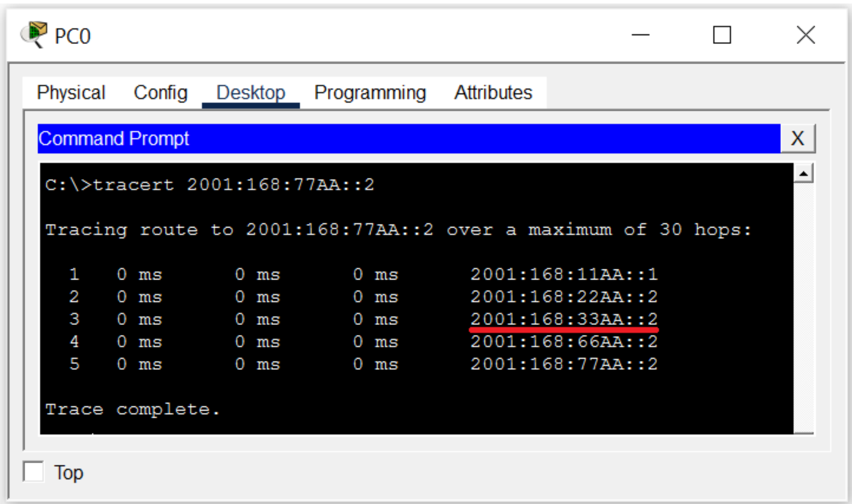
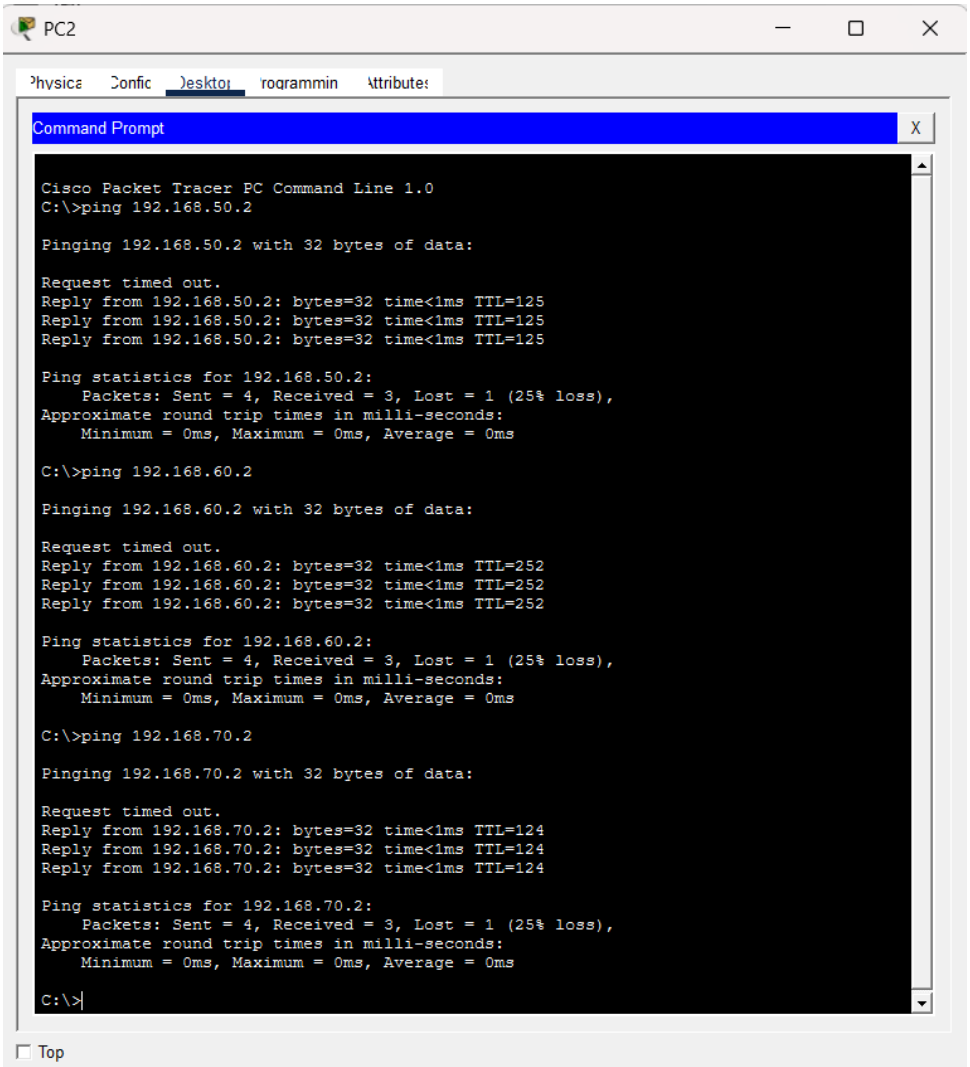


Figure 12: Traceroute from PC0 to a remote IPv6 network through the tunnel.

B) Ping from PC2 to Remote IPv4 Networks

To verify IPv4 connectivity, **PC2** sends *ICMP Echo Request* messages to IPv4 hosts located in remote parts of the network. This test validates that IPv4 routing paths across all routers are operational and correctly configured.

As shown in Figure 13, all *ICMP Echo Requests* receive corresponding *Echo Replies*. This confirms that IPv4 interfaces, IP addressing, and routing tables are properly configured, and that end-to-end IPv4 communication is functioning as expected.




```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.50.2

Pinging 192.168.50.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.50.2: bytes=32 time<1ms TTL=125
Reply from 192.168.50.2: bytes=32 time<1ms TTL=125
Reply from 192.168.50.2: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.50.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.60.2

Pinging 192.168.60.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.60.2: bytes=32 time<1ms TTL=252
Reply from 192.168.60.2: bytes=32 time<1ms TTL=252
Reply from 192.168.60.2: bytes=32 time<1ms TTL=252

Ping statistics for 192.168.60.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.70.2

Pinging 192.168.70.2 with 32 bytes of data:

Request timed out.
Reply from 192.168.70.2: bytes=32 time<1ms TTL=124
Reply from 192.168.70.2: bytes=32 time<1ms TTL=124
Reply from 192.168.70.2: bytes=32 time<1ms TTL=124

Ping statistics for 192.168.70.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 13: Successful IPv4 ping from PC2 to remote IPv4 networks.

C) HTTP Request from PC1 to the Web Server Using a Domain Name

PC1 is configured with IPv6-only addressing and initiates an HTTP request to the web server using its domain name. The DNS server is configured with both `A` (IPv4) and `AAAA` (IPv6) resource records for the web server.

As shown in Figure 14, the HTTP connection is successfully established over the IPv6 path. Because **PC1** has only an IPv6 address, the DNS resolution process selects the `AAAA` record, allowing **PC1** to reach the web server using IPv6 connectivity.

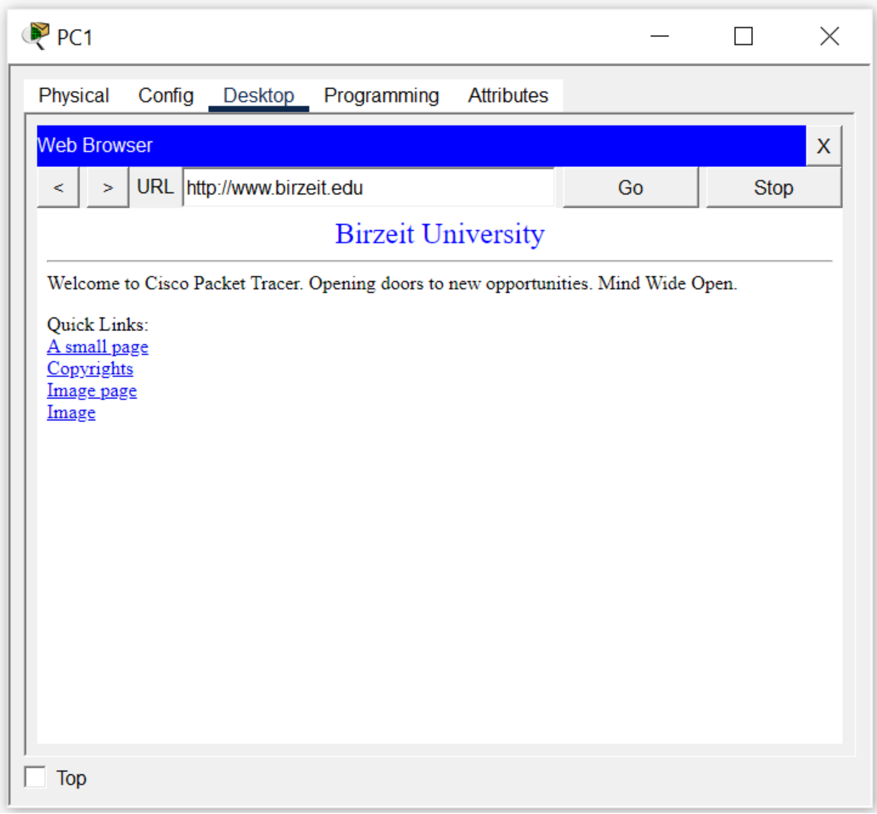


Figure 14: Successful HTTP request from PC1 using IPv6 and a domain name.

Switch Packet Tracer to Simulation Mode and initiate the HTTP request from **PC1**. Track all DNS query and response packets, the TCP connection establishment packets (three-way handshake), and the HTTP request and response exchange. Identify the ordering of the packets and determine which DNS record type is returned. Discuss your observations with the TA or Instructor, focusing on protocol selection and packet flow.

D) HTTP Request from PC3 to the Web Server Using a Domain Name

PC3 is configured as a dual-stack host, enabling it to communicate with the web server using either IPv4 or IPv6. When **PC3** initiates an HTTP request using the server's domain name, it first queries the DNS server, which returns both `A` (IPv4) and `AAAA` (IPv6) resource records.

Figure 15 shows that **PC3** successfully establishes an HTTP connection to the web server. Because IPv6 connectivity is available through the tunnel between **Router1** and **Router3**, **PC3** selects the IPv6 address provided by DNS. This behavior is consistent with standard dual-stack operation, in which IPv6 is preferred whenever it is reachable.

Using Simulation Mode in Packet Tracer, initiate communication from **PC3** and observe the complete process. Track the DNS queries and responses, noting the presence of both **A** and `AAAA` records. Then, follow the TCP handshake and subsequent HTTP packets to see which IP version is selected for the connection. Analyze the behavior of the system in handling both IPv4 and IPv6 addresses, highlighting the dual-stack support and the preference for IPv6 when both address types are available. Discuss these observations with the TA or Instructor to reinforce your understanding of IPv6 adoption and dual-stack networking.

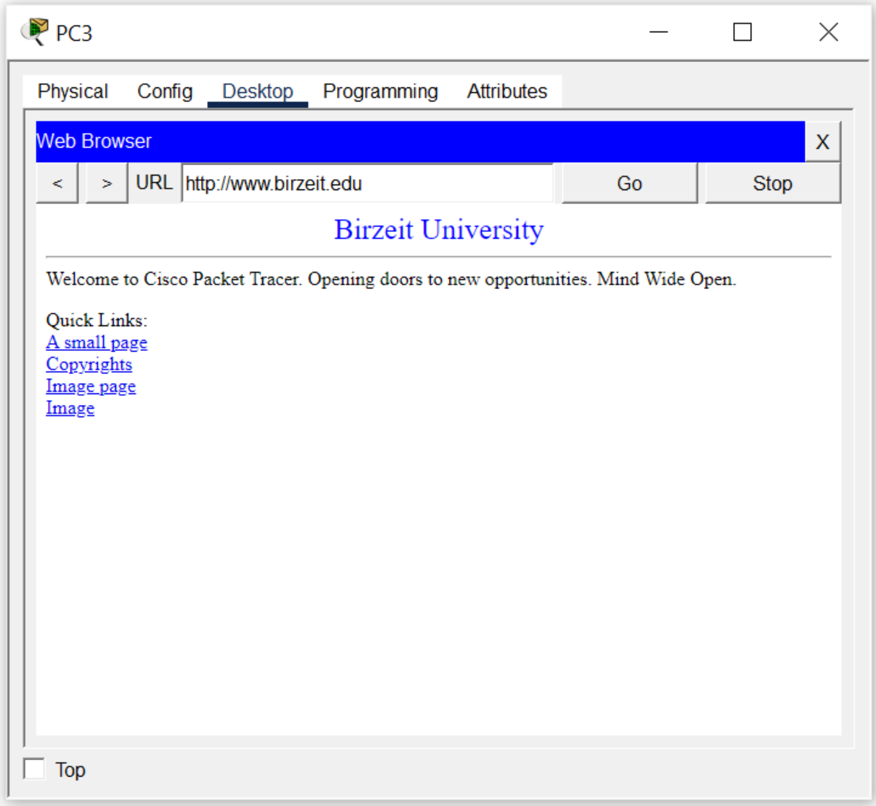


Figure 15: HTTP request from PC3 using a domain name.

10.4 ToDo

This section will be provided by the instructor.